

Critical-Degree M -Ideals for Finite-Dimensional Ranges

A partial answer to arXiv:1102.3850

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Model GPT5.6

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Status. Candidate partial result, likely valid. This packet solves the equality endpoint for every nonzero finite-dimensional range space F ; it does not settle the source question for arbitrary property- (M) spaces F .

1 Source question

The source is V. Dimant and S. Lassalle, *M -structures in vector-valued polynomial spaces*, published in Journal of Convex Analysis 19 (2012), 685–711 [2]. Its arXiv id is 1102.3850. On PDF page 22, immediately before Corollary 4.9, the authors state:

We do not know if the result remains true even for the case $n = cd(E, F) = cd(E)$.

Here the result under discussion is the following lifting statement. If $\mathcal{K}(E)$ is an M -ideal in $\mathcal{L}(E)$, F has property (M) , and

$$n = cd(E, F) < cd(E),$$

then $\mathcal{P}_w(^nE, F)$ is an M -ideal in $\mathcal{P}(^nE, F)$.

Now we can lift M -structures from the linear to the vector-valued polynomial context. This is done for the particular case of n -homogeneous polynomials when n is the critical degree of the pair (E, F) and it is strictly less than the critical degree of the domain space E . We do not know if the result remains true even for the case $n = cd(E, F) = cd(E)$.

Corollary 4.9. *Let E and F be Banach spaces and $n = cd(E, F) < cd(E)$. If $\mathcal{K}(E)$ is an M -ideal in $\mathcal{L}(E)$ and F has property (M) , then $\mathcal{P}_w(^nE, F)$ is an M -ideal in $\mathcal{P}(^nE, F)$.*

Proof. If $\mathcal{K}(E)$ is an M -ideal in $\mathcal{L}(E)$, appealing to [24] Theorem VI.4.17], E has property (M) and we may find $\{K_\alpha\}_\alpha \subset \mathcal{K}(E)$ a net of compact operators satisfying both conditions of Theorem 4.7. By Proposition 4.8 (E, F) has the n -polynomial property (M) . Now, we may apply Theorem 4.7 to derive the result. $\square$

2 Definitions and imported results

The critical degree $cd(E, F)$, when it exists, is the least positive integer k for which some continuous F -valued k -homogeneous polynomial on E is not weakly continuous on bounded sets. We write $cd(E) = cd(E, \mathbb{K})$.

A pair (E, F) has the n -polynomial property (M) when every $P \in \mathcal{P}(^n E, F)$ with $\|P\| \leq 1$ satisfies

$$\limsup_{\alpha} \|v + P(x_{\alpha})\| \leq \limsup_{\alpha} \|u + x_{\alpha}\|^n \quad (1)$$

whenever $\|v\| \leq \|u\|^n$ and (x_{α}) is a bounded weakly null net in E .

We use three results already proved in the cited papers.

- (i) If $\mathcal{K}(E)$ is an M -ideal in $\mathcal{L}(E)$, then E has property (M) and admits the compact approximation net appearing in Theorem 4.7 of [2].
- (ii) If $n = cd(E)$ and E has property (M) , then E has the scalar n -polynomial property (M) [1, Proposition 3.10].
- (iii) At $n = cd(E, F)$, the compact approximation hypotheses in (i) and the vector n -polynomial property (M) imply that $\mathcal{P}_w(^n E, F)$ is an M -ideal in $\mathcal{P}(^n E, F)$ [2, Theorem 4.7].

3 Proof intuition

The only new issue is lifting a scalar property- (M) inequality to a vector norm. In finite dimensions a norm can be scalarized coherently along the net: choose a norming functional at each point, then pass to a subnet on which those functionals converge in norm. The moving scalarizations can therefore be replaced by one fixed functional f , reducing the desired inequality to scalar property (M) for $f \circ P$.

This mechanism is unavailable in general infinite-dimensional ranges: weak-star convergence of norming functionals does not control their diagonal pairing with the moving vectors $P(x_{\alpha})$.

4 Formal proof

Lemma 1. *If $F \neq \{0\}$ is finite-dimensional and $cd(E)$ exists, then*

$$cd(E, F) = cd(E).$$

Proof. Fix $0 \neq y \in F$. If a scalar k -homogeneous polynomial p is not weakly continuous on bounded sets, neither is $x \mapsto p(x)y$. Thus $cd(E, F) \leq cd(E)$.

Conversely, let $k < cd(E)$ and $P \in \mathcal{P}(^k E, F)$. For every $f \in F^*$, the scalar polynomial $f \circ P$ is weakly continuous on bounded sets. Choose a finite basis $f_1, \dots, f_m$ of F^* . If $x_{\alpha} \rightarrow x$ weakly in a bounded subset of E , then $f_j(P(x_{\alpha}) - P(x)) \rightarrow 0$ for each j . Coordinate convergence in finite-dimensional F is norm convergence, so $P(x_{\alpha}) \rightarrow P(x)$ in norm. Hence every such P is weakly continuous on bounded sets and $cd(E, F) \geq cd(E)$. $\square$

Lemma 2. *Let F be finite-dimensional. If E has scalar n -polynomial property (M) , then (E, F) has vector n -polynomial property (M) .*

Proof. Let $P \in \mathcal{P}(^n E, F)$ have norm at most one. Fix $u \in E$, $v \in F$ with $\|v\| \leq \|u\|^n$, and a bounded weakly null net (x_{α}) . Put

$$A = \limsup_{\alpha} \|v + P(x_{\alpha})\|, \quad B = \limsup_{\alpha} \|u + x_{\alpha}\|^n.$$

Pass to a subnet, still denoted (x_α) , on which the first norm converges to A . For every α , Hahn–Banach gives $f_\alpha \in S_{F^*}$ such that

$$f_\alpha(v + P(x_\alpha)) = \|v + P(x_\alpha)\|. \quad (2)$$

This choice is valid over both real and complex scalars.

The unit sphere of F^* is norm compact. After passing to a further subnet, $f_\alpha \rightarrow f$ in norm for some $f \in B_{F^*}$. The net $(v + P(x_\alpha))$ is bounded, so (2) implies

$$|f(v) + (f \circ P)(x_\alpha)| \rightarrow A.$$

The scalar polynomial $f \circ P$ has norm at most one and

$$|f(v)| \leq \|v\| \leq \|u\|^n.$$

Scalar n -polynomial property (M) therefore yields

$$A \leq \limsup_\alpha \|u + x_\alpha\|^n \leq B.$$

The last inequality holds because the selected net is a subnet of the original net. This is exactly (1). $\square$

Theorem 3 (Finite-dimensional endpoint). *Let E be a Banach space whose critical degree $n = cd(E)$ exists, and let $F \neq \{0\}$ be finite-dimensional. If $\mathcal{K}(E)$ is an M -ideal in $\mathcal{L}(E)$, then*

$$\mathcal{P}_w(^n E, F) \text{ is an } M\text{-ideal in } \mathcal{P}(^n E, F).$$

In particular, Corollary 4.9 of [2] remains true at $n = cd(E, F) = cd(E)$ for every finite-dimensional F .

Proof. The M -ideal hypothesis on $\mathcal{K}(E)$ gives property (M) of E and the compact approximation net required by [2, Theorem 4.7]. By [1, Proposition 3.10], E has scalar n -polynomial property (M) . Lemma 2 upgrades this to vector n -polynomial property (M) for (E, F) , while Lemma 1 gives $n = cd(E, F)$. Applying [2, Theorem 4.7] proves the conclusion. $\square$

5 Deep upgrade attempt and limitation

The finite-dimensional proof suggests choosing norming functionals f_α for an arbitrary property- (M) range F . Banach–Alaoglu only gives a weak-star convergent subnet. It does not imply

$$(f_\alpha - f)(P(x_\alpha)) \rightarrow 0,$$

because both factors move with α , and at the critical degree $P(x_\alpha)$ need not be weakly null. This is exactly the point at which the strict inequality $n < cd(E)$ was used in Proposition 4.8 of [2].

I also tested compact-operator approximation on F . A fixed compact operator makes the norming-functional argument finite-dimensional on its image, but no such operator uniformly approximates the range of an arbitrary critical-degree polynomial. The stronger M_∞ -decomposition estimate does handle important infinite-dimensional targets, but that is already the mechanism of Propositions 2.7–2.8 in [2]. No general property- (M) upgrade is claimed.

6 Verification and novelty status

The proof uses no computation. The separate `verification.md` audits critical-degree equality, complex Hahn–Banach scalarization, limsup subnets, and every imported theorem.

Local indexes were searched for arXiv:1102.3850, the exact title, n -polynomial property (M), the equality $cd(E, F) = cd(E)$, and finite-dimensional ranges. Bounded web searches used the same exact phrases and checked the source, the scalar predecessor arXiv:1005.1260, and the later related arXiv:1512.08741. No exact finite-dimensional endpoint statement was found. This is bounded evidence, not a guarantee of novelty.

Human-review recommendation

Promote as a likely-valid partial result after checking the two-line subnet scalarization in Lemma 2 and the cited theorem numbers. The main claim should remain explicitly limited to finite-dimensional F .

References

- [1] V. Dimant, *M-ideals of homogeneous polynomials*, Studia Math. 202 (2011), 81–104; arXiv:1005.1260.
- [2] V. Dimant and S. Lassalle, *M-structures in vector-valued polynomial spaces*, J. Convex Anal. 19 (2012), 685–711; arXiv:1102.3850.